

Why 192.168 used with so many home use routers

Because **192.168.x.x** is specifically reserved for private networks**, and home-router manufacturers adopted it as a convenient default.

The key comes from **RFC 1918****, which defines three IPv4 address ranges that anyone can use internally without conflicting with addresses on the public Internet:

Private range	CIDR	Approx. addresses
-----	-----	-----:
10.0.0.0 – 10.255.255.255	10.0.0.0/8	16.7 million
172.16.0.0 – 172.31.255.255	172.16.0.0/12	1 million
192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536

Why 192.168 became the home-network favorite

A typical home network doesn't need thousands or millions of addresses. So manufacturers commonly carve out a `/24` network from 192.168, such as:

```
```text
192.168.0.0/24
192.168.1.0/24
192.168.50.0/24
```
```

For example, with:

```
```text
192.168.1.0/24
```
```

you'll commonly see:

```
```text
192.168.1.1 Router/default gateway
192.168.1.2 Computer
192.168.1.3 Printer
192.168.1.4 Smart TV
...
192.168.1.254
```
```

The router normally runs **DHCP**** to hand these addresses to devices automatically.

But how can millions of houses use 192.168.1.1?

That's the important part for cybersecurity students: **those addresses aren't globally unique.**

Your computer might be:

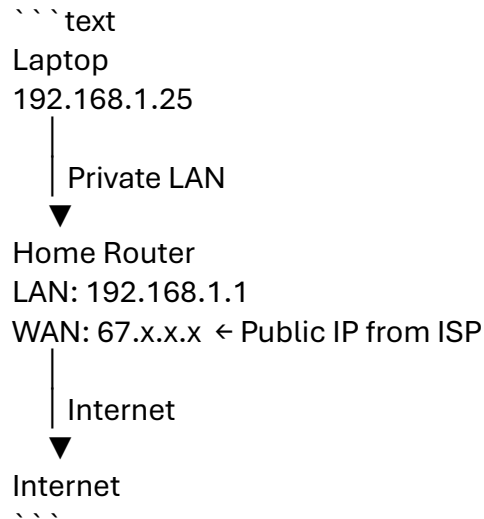
192.168.1.25

while thousands of other computers around the country are simultaneously

192.168.1.25.

That's okay because those addresses exist only within their respective private networks. Your router uses **NAT (Network Address Translation)** to translate traffic between private addresses and the public IP address assigned by the ISP.

A simplified picture is:



That's also why the Nmap example:

```
```bash
nmap -sn 192.168.1.0/24
```
```

is so common in labs. It's essentially saying:

> **"Find the devices on my local 192.168.1.x network."**

One teaching point worth emphasizing: **not every 192.x.x.x address is private.** Only **192.168.0.0 through 192.168.255.255** is RFC 1918 private space. Similarly, with 172, only **172.16–172.31** is private—not all 172 addresses.